

Setting Up FreeCAD 1.1

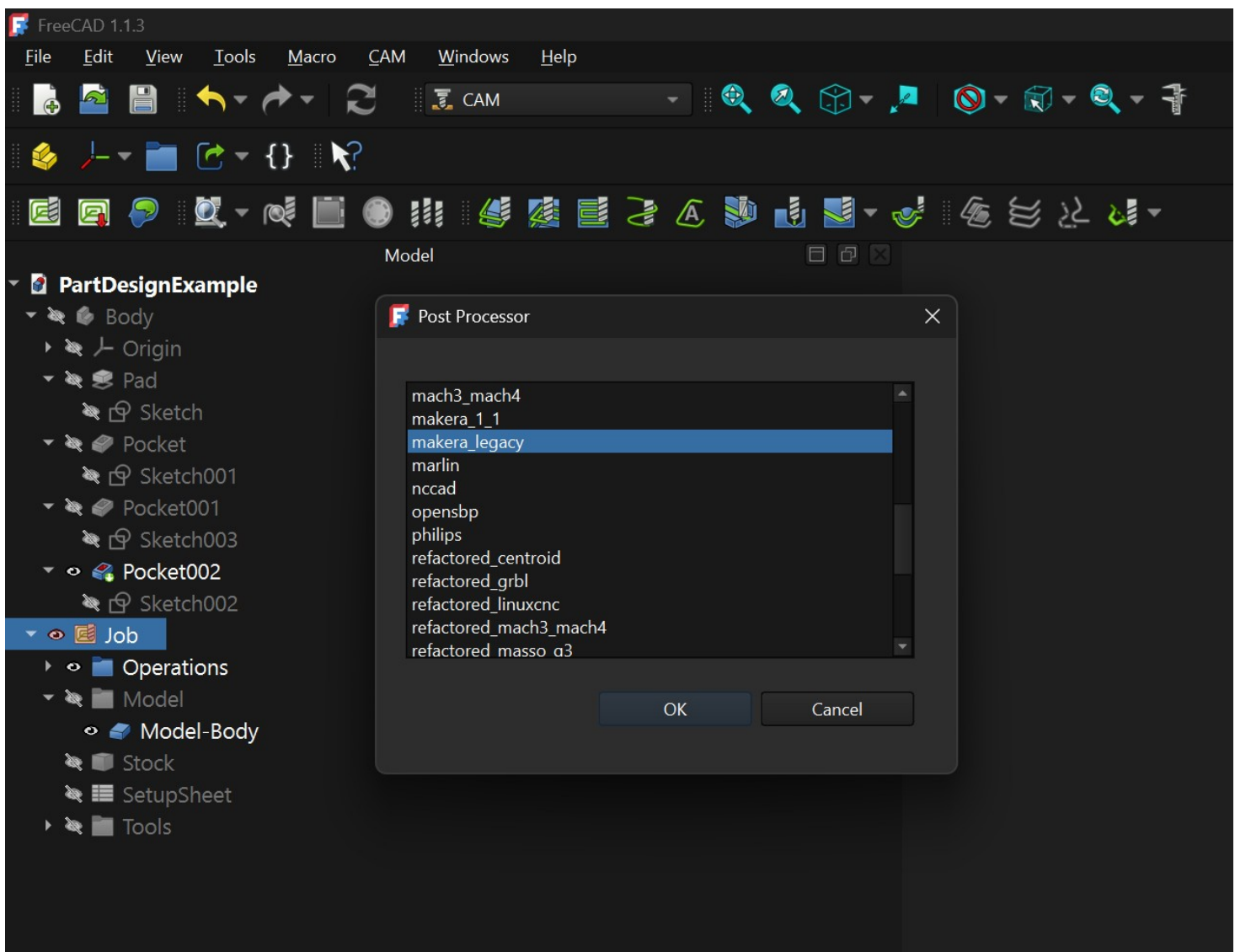
On Windows:

- Copy the makera_legacy_post.py to your freecad install directory. By default this is:
C:\Program Files\FreeCAD 1.0\Mod\CAM\Path\Post\scripts

• On linux:

- Copy the makera_legacy_post.py to the path found in the freecad Preferences>Python>Macro>Macro Path

- Then just set "makera_legacy" as the post-processor in the CAM workbench.



Setting Up FreeCAD 26.3 (also called 1.2) and beyond

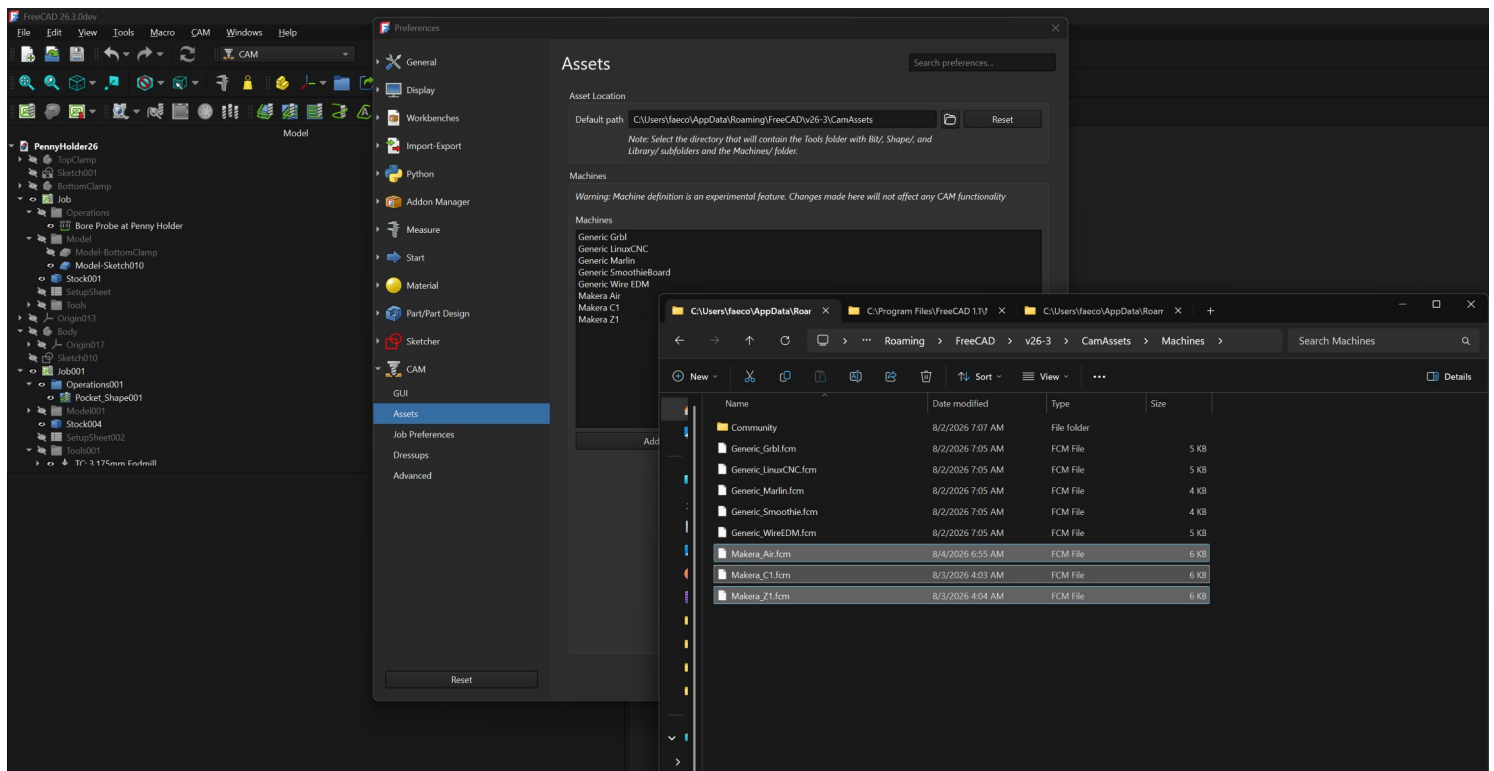
You can use the same legacy post processor as 1.1 for now, however, the machine based post processor offers a few improvements and the legacy post processor will get removed from the software at some point

On Windows:

- Copy the makera_26_MCH_post.py to your freecad install directory. By default this is:
C:\Program Files\FreeCAD 1.0\Mod\CAM\Path\Post\scripts

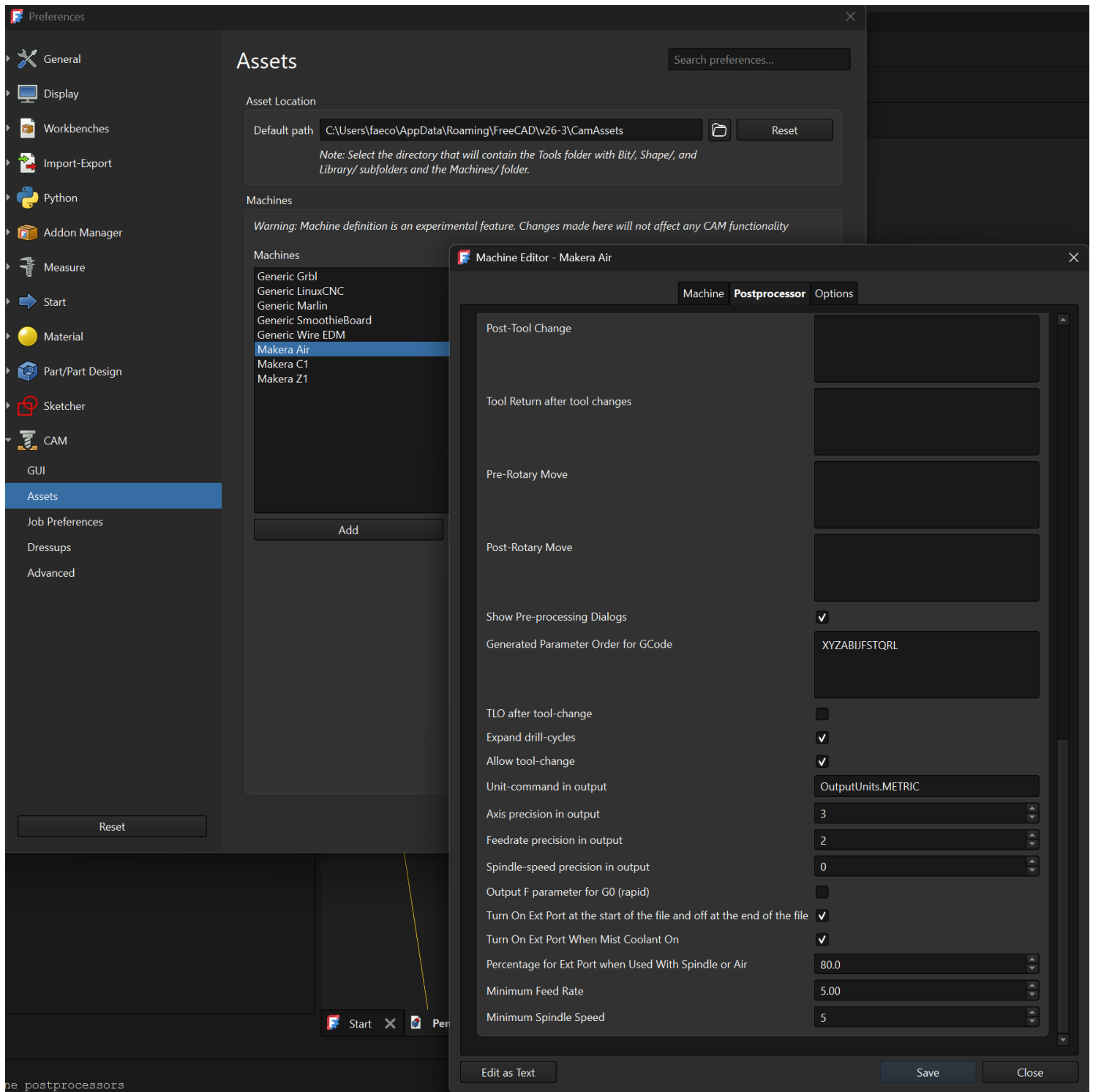
• On linux:

- add the file to the path found in the freecad Preferences>Python>Macro>Macro Path
- Enter the CAM workspace and open preferences - > CAM - > Assets
- Copy the .fcm machines to a subfolder Machines under this path



Setting Up Freecad 26.3 Pt. 2

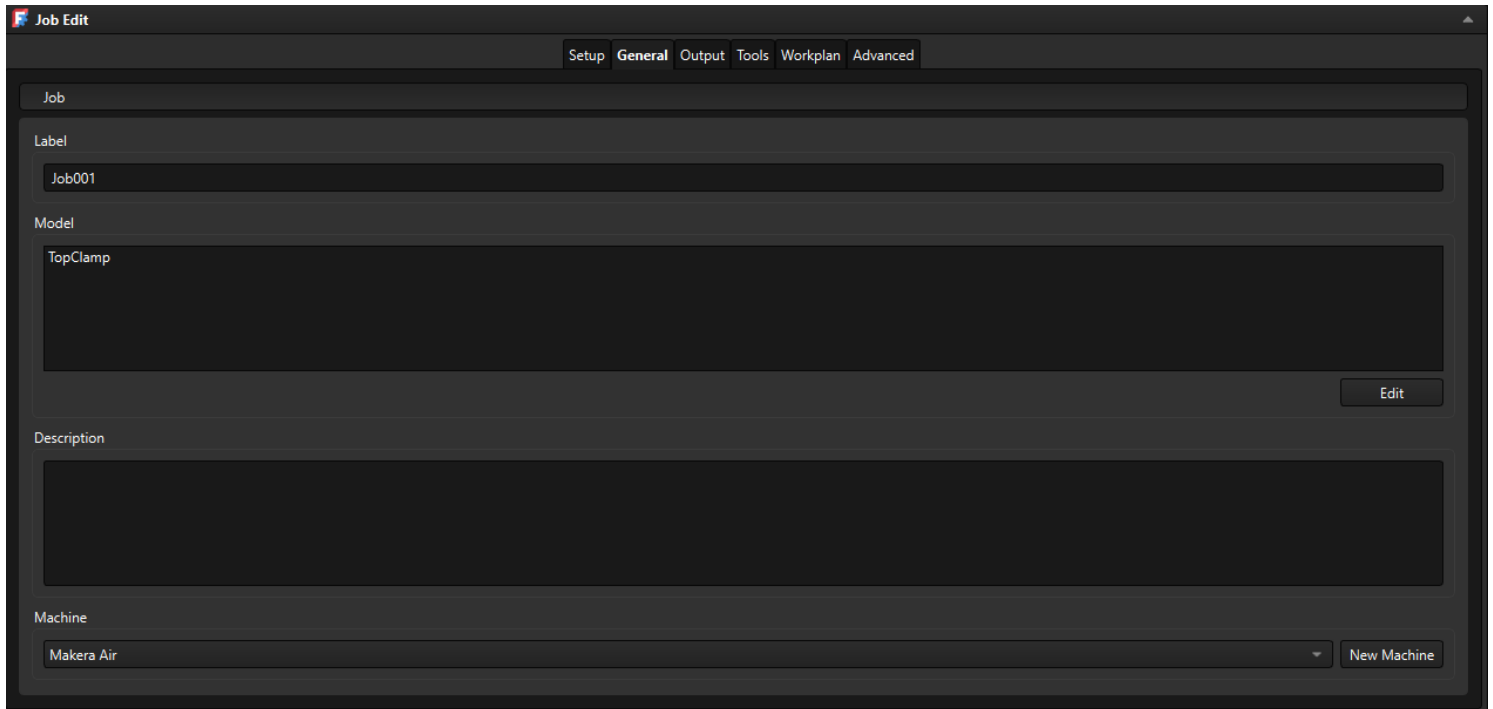
Here you can also edit the post processor settings to turn on the EXT port at the start and end of the file, or every time a tool is set to use mist coolant. The rest of the settings you should generally leave alone



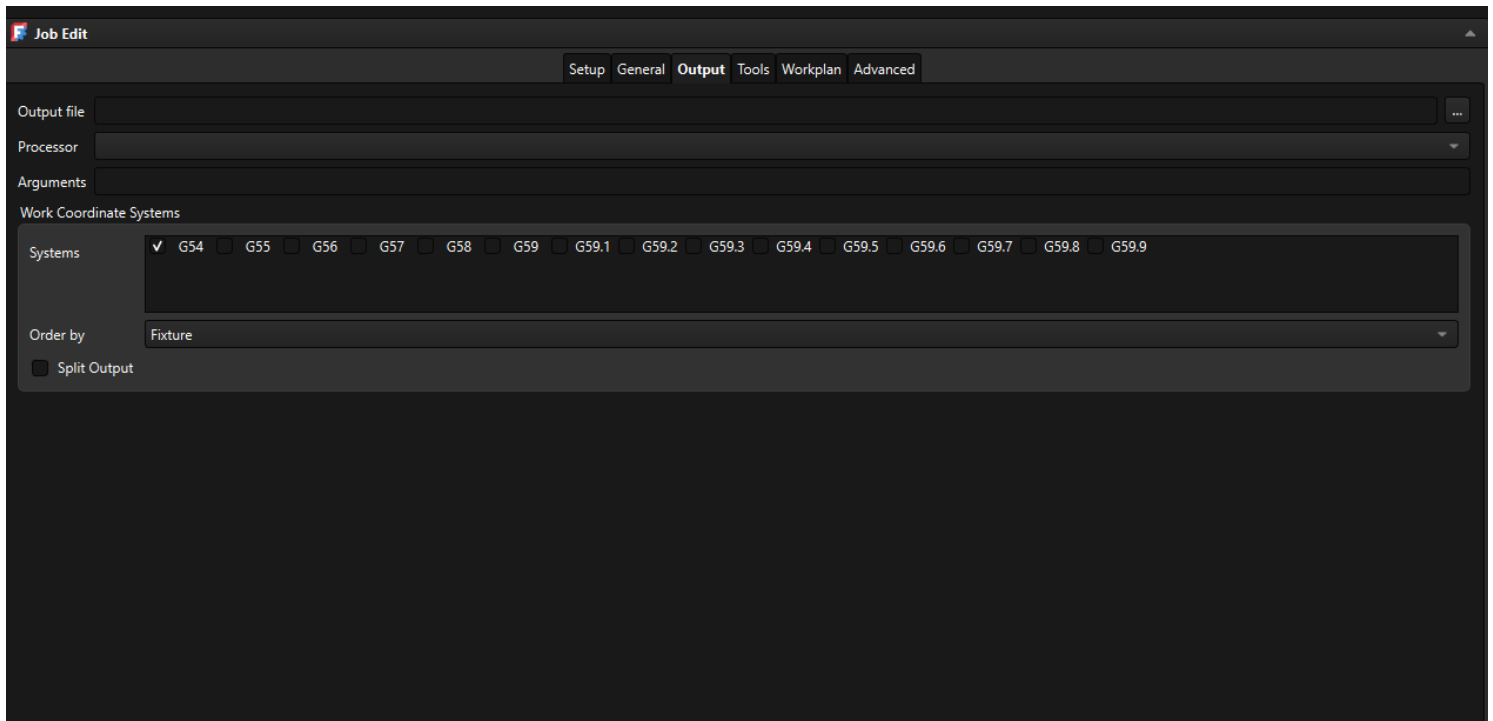
Setting Up Freecad 26.3 Pt. 3

Here you can also edit the post processor settings to turn on the EXT port at the start and end of the file, or every time a tool is set to use mist coolant.
The rest of the settings you should generally leave alone

To use the machine post processor, when you set up a job select the machine in the general tab

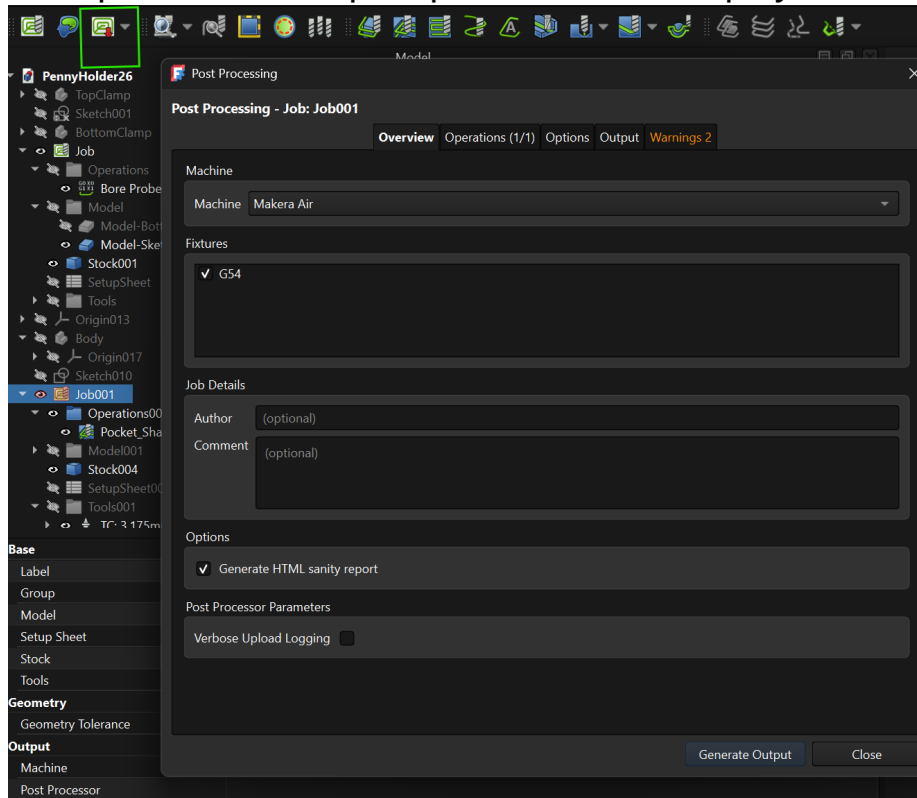


And leave the post processor file blank in the output tab

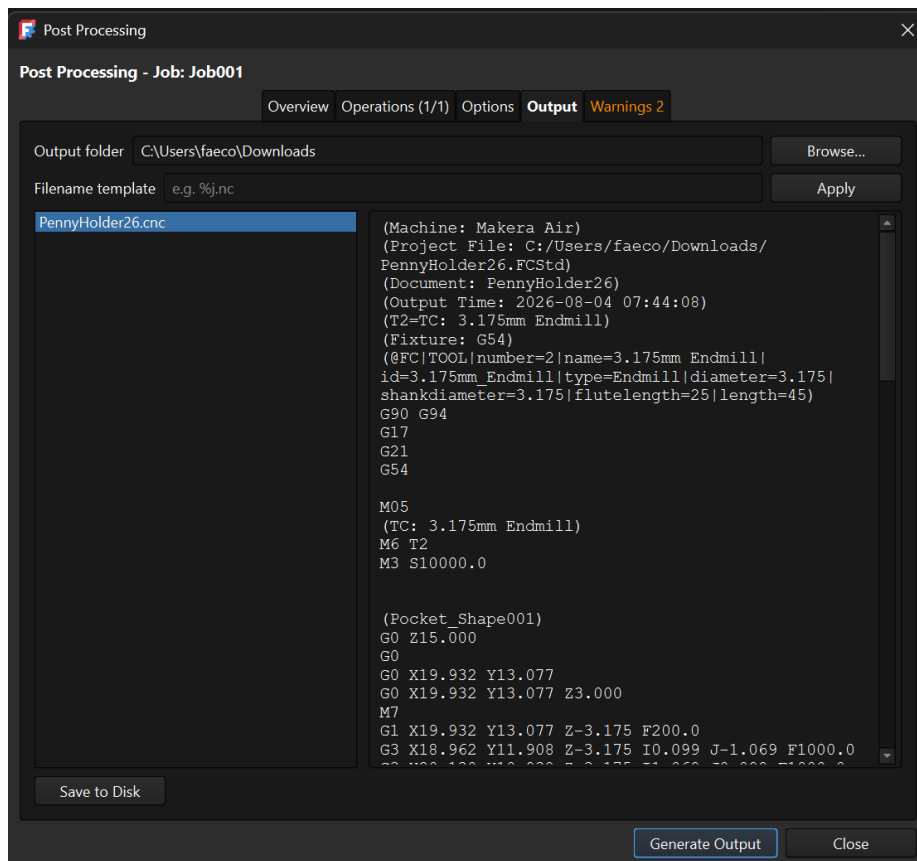


Setting Up Freecad 26.3 Pt. 4

Now when you post process that job you will have a post process popup
Clicking generate output will run the post processor and display it in the output tab

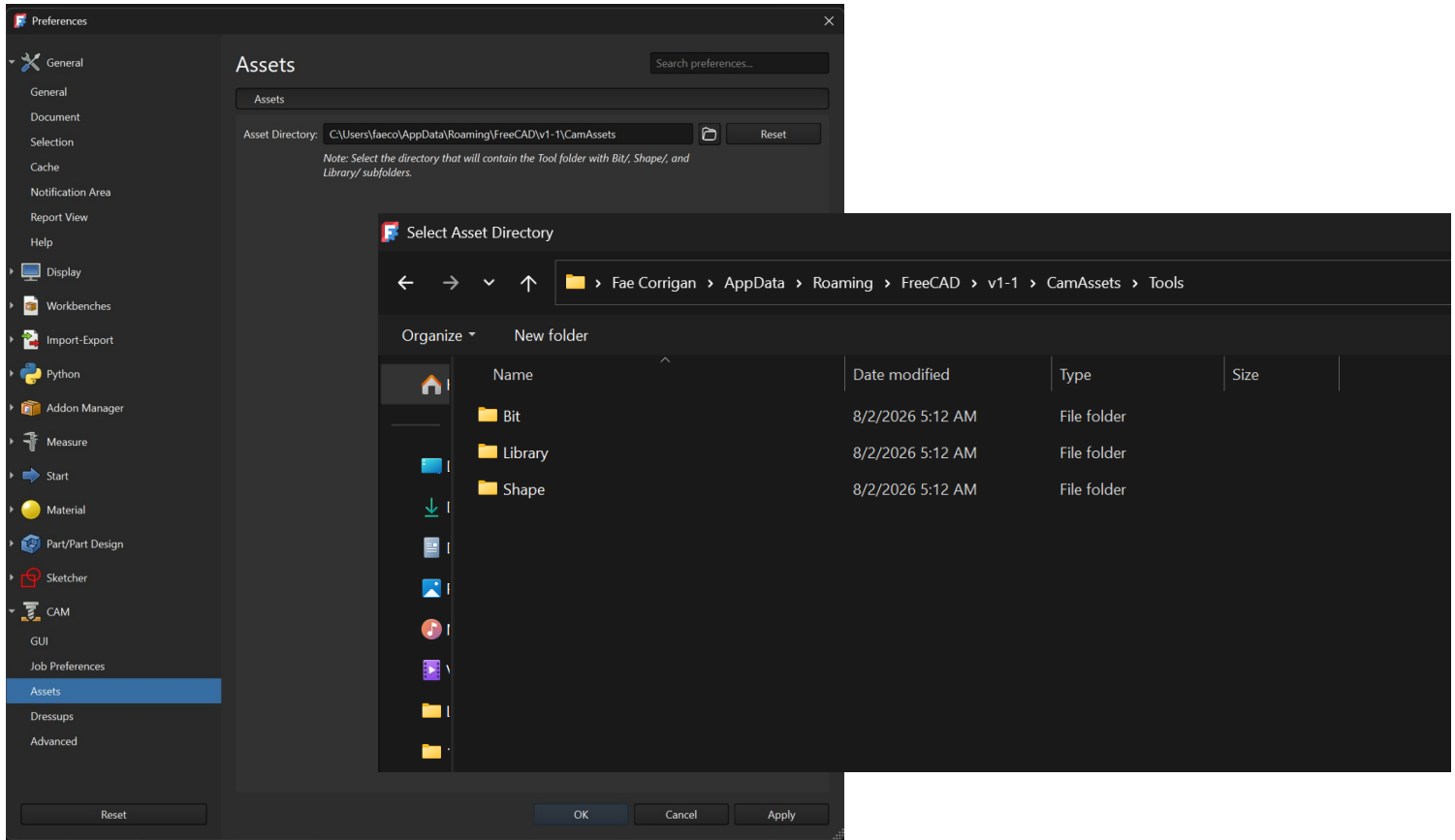


And you can save the output to disk in the output tab

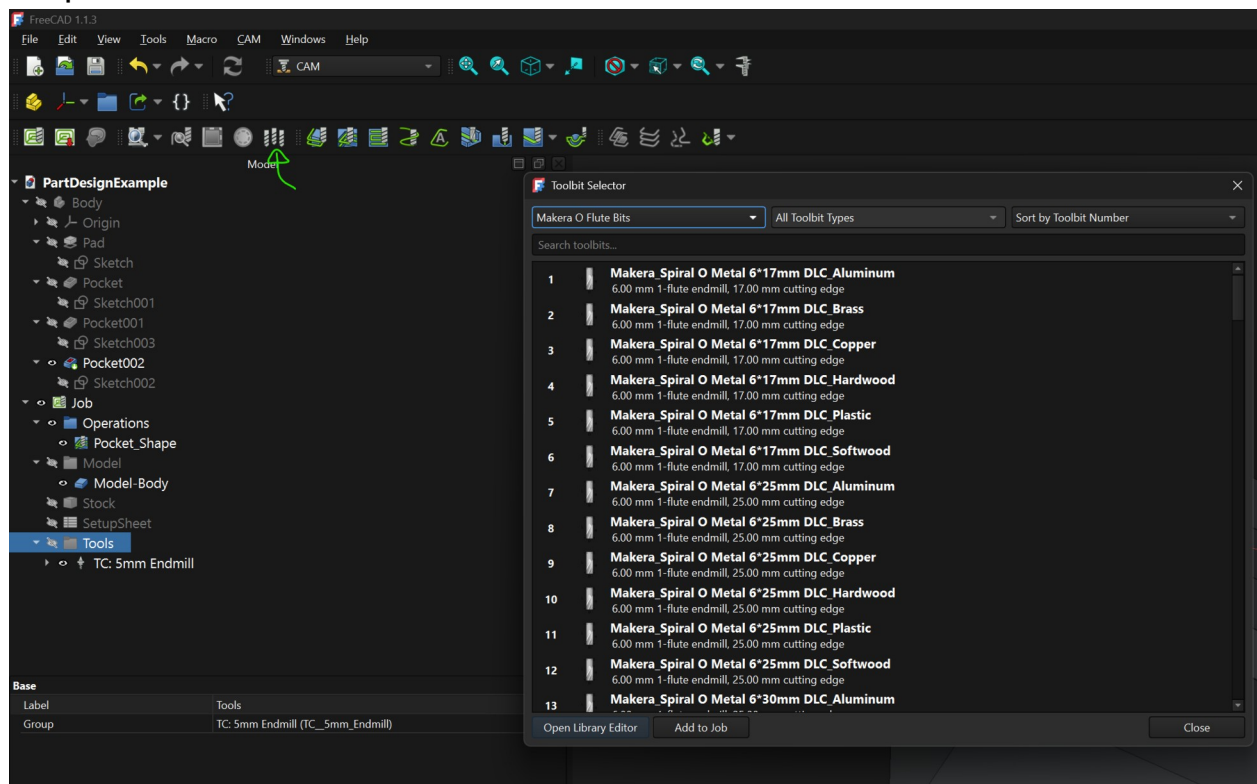


Setting Up Freecad Tools (1.1 and 26.3+)

Add the tool files into the directory found in the freecad preferences



Then add the tools to your project in the usual way and there will be makera categories in the dropdowns



After you have added a tool into the project you need to modify its feeds and speeds by double clicking the tool modifying the properties. Here is also where you can set the toolnumber for the tool as it will be used on this project.

PartDesignExample

Body

Origin

Pad

Sketch

Pocket

Sketch001

Pocket001

Sketch003

Pocket002

Sketch002

Job

Operations

Pocket_Shape

Model

Model-Body

Stock

SetupSheet

Tools

TC: 5mm Endmill

TC: Makera_Spiral O Metal 6*17mm DLC_Aluminum

Tasks

OK

Cancel

Tool Controller Editor

Controller Name / Tool Number

iral O Metal 6*17mm DLC_Aluminum

1

Horizontal feed

0.00 mm/min

fx

Vertical feed

0.00 mm/min

fx

Horizontal rapid

0.00 mm/min

fx

Vertical rapid

0.00 mm/min

fx

Spindle

0.00

Forward

Base

Label

TC: Makera_Spiral O Metal 6*17mm DLC_Aluminum

Tool

Makera_Spiral O Metal 6*17mm DLC_Aluminum (Makera_Spiral_O_Metal_6_17...

Feed

Horiz Feed

0.00 mm/min

Vert Feed

0.00 mm/min

Rapid

Horiz Rapid

0.00 mm/min (SetupSheet.HorizRapid)

Vert Rapid

0.00 mm/min (SetupSheet.VertRapid)

Tool

Spindle Dir

Forward

Spindle Speed

0.00

Tool Number

1

You can also modify the tool feeds and speeds per operation by editing the operation and selecting edit tool controller

PartDesignExample

- Body
 - Origin
 - Pad
 - Sketch
 - Pocket
 - Sketch001
 - Pocket001
 - Sketch003
 - Pocket002**
 - Sketch002
- Job
 - Operations
 - Pocket_Shape**
 - Model
 - Model-Body
 - Stock
 - SetupSheet
 - Tools
 - TC: 5mm Endmill
 - TC: Makera_Spiral O Metal 6*17mm DLC_Aluminum

Base

Placement	[(0.00 0.00 1.00); 0.00°; (0.00 mm 0.00 mm 0.00 mm)]
Label	Pocket_Shape

Depth

Clearance Height	6.00 mm (OpStockZMax + SetupSheet.ClearanceHeightOffset)
Final Depth	-50.00 mm (OpFinalDepth)
Finish Depth	0.00 mm
Safe Height	4.00 mm (OpStockZMax + SetupSheet.SafeHeightOffset)
Start Depth	1.00 mm (OpStartDepth)
Step Down	5.00 mm (OpToolDiameter)

Extension

Extension Corners	☒ Yes
Extension Length Default	2.50 mm (OpToolDiameter / 2)

Op Values

Op Final Depth	-50.00 mm
Op Start Depth	1.00 mm
Op Stock ZMax	1.00 mm
Op Stock ZMin	-101.00 mm
Op Tool Diameter	5.00 mm

Path

Active	☒ Yes
Base	Model-Body (Clone) [Face23]
Comment	
Coolant Mode	None
Cycle Time	Tool Feedrate Error
Locations	[]
Split Arcs	☐ No
Tool Controller	TC: 5mm Endmill (TC_5mm_Endmill)
User Label	

Tasks

OK Cancel Apply

Pocket Shape

Base Geometry Extensions Depths Heights Operation

Tool controller TC: 5mm Endmill

Coolant Mode None

☒ Edit Tool Controller

This tool controller is used by 0 other operations.

Controller Name / Tool Number

TC: 5mm Endmill 1

Horizontal feed 0.00 mm/min

Vertical feed 0.00 mm/min

Horizontal rapid 0.00 mm/min

Vertical rapid 0.00 mm/min

Spindle

0.00 Forward

Cut Mode Climb

Pattern Offset

Angle 45.00°

Step over percent 50

Pass Extension 0.00 mm

☐ Use start point ☐ Use outline

☐ Min travel

☐ Use rest machining